

1. Evaluate the following integrals by integration by parts.

1).  $\int x e^{-x} dx$

2).  $\int t \sin(2t) dt$

3).  $\int \ln(2x + 1) dx$

4).  $\int \sin^{-1} x dx$

$$5). \int_0^{\frac{\pi}{6}} t \cos(3t) dt$$

$$6). \int_1^2 \frac{\ln x}{x^2} dx.$$

$$7). \int x^4 (\ln x) dx$$

$$8). \int e^{2\theta} \sin(3\theta) d\theta.$$